LLMs can reason if we consider the alternative decoding paths. Contrastingly, an intriguing phenomenon emerges when exploring alternative top-k ($k > 0$) tokens at the *first decoding step*. Continuing with greedy decoding from this point reveals natural CoT reasoning in many cases. These findings suggest that large language models possess inherent reasoning capabilities for numerous tasks following pre-training, but these abilities are obscured by the predominant use of greedy decoding. These reasoning paths can be easily uncovered by incorporating alternative decoding paths.

For instance, in the GSM8K question (Table 1), a valid CoT emerges at $k = 9$. Similarly, in the year parity task, greedy decoding attempts to directly answer the parity question at $k = 0$, leading to a random choice between "even" and "odd" which often results in an incorrect answer. However, when exploring $k > 0$, the model naturally generates CoT paths at $k = 3$ and $k = 7$, where it first determines the year before resolving the parity.

2.2. CoT-Decoding for Extracting CoT Paths

In this section, we further show how we can reliably extract those CoT-paths during the decoding process. Table 1 illustrates that CoT paths do not consistently outrank non-CoT ones in the model's probability assessment. Moreover, they often do not represent the predominant answer among all paths, rendering methods like self-consistency (Wang et al., 2023a) inapplicable. For instance, in the GSM8K question, the prevalent answer "60", which aligns with the greedy decoding result, fails to serve as a reliable indicator for identifying the correct path.

Interestingly, upon examining the model's logits, we found that the presence of a CoT path typically leads to a more *confident* decoding of the *final answer*, characterized by a significant probability disparity between the top and secondary tokens:

$$\Delta_{k,\text{answer}} = \frac{1}{|\text{answer}|} \sum_{x_t \in \text{answer}} p(x_t^1 \mid x_{<t}) - p(x_t^2 \mid x_{<t}).$$

Here x_t^1 and x_t^2 represent the top two tokens at the t-th decoding step in the k-th decoding path, chosen for their maximum post-softmax probabilities from the vocabulary, given x_t being part of the answer tokens. This uncertainty measure is similar to the minimum-margin approach in (Jiang and Gupta, 2019) and in our case, the model's overall confidence in decoding the final answer is approximated by averaging these probability differences for all relevant answer tokens x_t. For example, for the GSM8K question in Table 1, given the answer "60", we average the probability differences for all tokens in that answer, i.e., "6" and "0".[2]

This method, referred to as **CoT-decoding**, extracts such CoT paths among the decoded paths from the model. As illustrated in Table 1, each decoding path is marked with its corresponding Δ value in blue (the answer tokens are bolded). It is evident that paths with a CoT component exhibit a significantly higher Δ, highlighting the model's increased confidence, as opposed to paths without CoT. We also did a quantitative analysis by manually examining the first 100 questions in GSM8K, and among those, if we take the decoding path with the highest answer confidence among the top-10 decoding paths, 88% of them contain CoT paths. This shows an overwhelmingly high correlation between the model's answer confidence and the CoT paths.

Comparing different CoT-path extraction approaches. In Table 2, we compare different ways to extract the CoT-paths out of the top-10 decoded paths. It is easy to see that the model's own

[2]We also considered other popular choices for measuring the model's uncertainty (Settles, 2009), e.g., using the model's probability on the token itself (i.e., $p(x_t^1 \mid x_{<t})$ only), which performs slightly worse compared to the min-margin approach. In addition, an entropy estimate is not accurate due to the large vocabulary size in LLMs and the common use of vocabulary truncation.

probability measure does not serve as a reliable indicator, nor does the model's length-normalized probability (since an intuition could be a CoT-path should usually be a longer decoding path, which is not always the case, e.g., on the year parity task). In contrast, CoT-decoding can reliably extract the CoT-paths, yielding a significant boost on the model's reasoning performance.

	GSM8K (top-100)	Year Parity
Greedy decoding	44.0%	57.0%
Decode 10 paths, rank by model's highest log-prob	37.0%	55.0%
Decode 10 paths, rank by model's highest length-normalized log-prob	51.0%	57.0%
CoT-decoding (decode 10 paths, rank by model's answer confidence)	**72.0%**	**95.0%**

Table 2 | CoT-decoding reliably extracts the CoT-paths compared to other methods (on PaLM-2 L).

Identify the answer spans. Computing Δ requires identifying the answer spans in a model's response. One common approach used for public models is to extract the last numerical value in math reasoning tasks, or the final option in set-based reasoning tasks, as the answer, following the Tülu evaluation (Ivison et al., 2023; Liu et al., 2024; Wang et al., 2023b). Alternatively, similarly to the method used in Kojima et al. (2022), we can also extend the model's output with the prompt "So the answer is", and then align these continuations with spans in the model's decoding path as the answer.

Sampling under the standard QA format. CoT-decoding explores alternative tokens at the first decoding step. A natural question arises: can sampling achieve a similar effect and unveil the CoT reasoning paths? We found that, although sampling works well under few-shot CoT prompting (Wang et al., 2023a), it does not exhibit the desired behaviour without the prompts. We compare CoT-decoding with self-consistency when no CoT prompt is used in Table 3. The ineffectiveness of sampling stems from the model's strong tendency in providing a direct answer during decoding, hence the first token tends to have less diversity compared to CoT-decoding. In contrast, CoT-decoding works by explicitly encouraging diversity at the first decoding step.

Table 3 | CoT-decoding and self-consistency w/o prompts on GSM8K.

	Mistral-7B	PaLM-2 L
Greedy decoding	9.9%	34.8%
Self-consistency without CoT-prompt (10 paths)	12.9%	40.6%
CoT-decoding (10 paths)	**25.1%**	**63.2%**

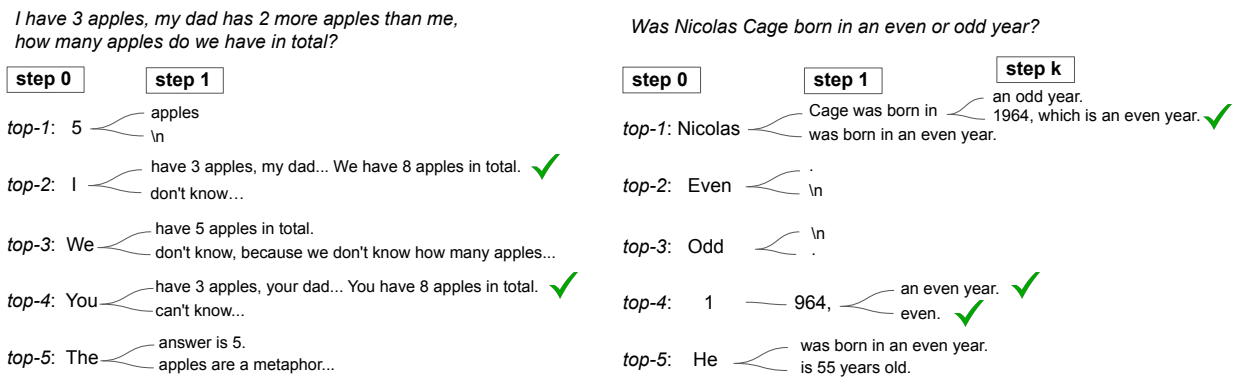

Figure 2 | Decoded paths by considering alternative tokens at various decoding steps.

Branching at other decoding steps. Another natural question is whether branching is viable at later decoding stages, comparing to only branching at the first decoding step. In Figure 2, we highlight the impact of alternative token consideration in subsequent decoding steps. It is evident that early branching, e.g., at the first decoding step, significantly enhances the diversity of potential paths.

Conversely, later-stage branching is significantly influenced by previously generated tokens. For instance, initiating with the token "5" greatly decreases the likelihood of rectifying an erroneous path. Nonetheless, the optimal branching point may vary with the task; in the year parity task, for instance, mid-path branching can effectively yield correct CoT paths.

Aggregation of the decoding paths. Since we already decode the top-k paths, one natural extension is to aggregate the answers over all those paths, similar to self-consistency (Wang et al., 2023a) but without the use of prompts. The rationale behind this aggregation is to mitigate sensitivity to small differences in the model's logits, particularly when relying solely on the path with the maximum Δ. The examples in Table 1 show that the majority answer is unlikely to be the correct one. Instead, we propose a weighted aggregation method, i.e., we take the answer that maximizes $\tilde{\Delta}_a = \sum_k \Delta_{k,a}$ where $\Delta_{k,a}$ is the k-th decoding path whose answer $= a$. We found that adopting this approach enhances the stability of the results, and further analysis is presented in Section §3.3.

3. Experiments

Experiment Setup. For all experiments, the default input to the model is the standard QA format of *Q: [question]\nA:*, where *[question]* is filled with the actual question depending on the task, and we ask the model to continue the generation given that prefix. During decoding, we use $k = 10$ as default for the alternative top-k tokens at the first decoding position, and continue greedy decoding afterwards. We show ablation studies with respect to the different choice of k in Section §3.1.

Datasets. For mathematical reasoning, we use the Grade-school math problems (GSM8K; Cobbe et al., 2021a) and the multi-step arithmetic dataset from (MultiArith; Roy and Roth, 2015). For commonsense reasoning, we investigate the "year parity" task where recent literature finds large language models still struggle with. The task is to query the model with "Was [person] born in an even or odd year?" where "[person]" is filled by a random celebrity name.[3] Existing work (Allen-Zhu and Li, 2023; Berglund et al., 2023) shows that even SoTA models like GPT-4 struggle with such tasks, achieving at-chance accuracy (~50%) when prompted directly. Additionally, we investigate symbolic reasoning tasks from Big-Bench-Hard (bench authors, 2023; Suzgun et al., 2022).

Models. We use three public models: (1) PaLM-2 (Anil et al., 2023) with different scales, ranging from X-Small, Small, Medium, and Large; (2) Mistral-7B (Jiang et al., 2023), and (3) Gemma-7B (Team et al., 2024). Our experiments primarily focus on pre-trained models, but we also include experiments with instruction-tuned models (denoted as "inst-tuned" or "IT").

3.1. CoT-Decoding Effectively Elicits Reasoning from Language Models

CoT-decoding is the only decoding strategy that effectively improves language model reasoning. In Table 4, we present results from popular decoding baselines on the Mistral-7B pre-trained model, including temperature sampling (Ackley et al., 1985; Ficler and Goldberg, 2017), top-k sampling (Fan et al., 2018; Holtzman et al., 2018; Radford et al., 2019), nucleus sampling (Holtzman et al., 2020), and beam search. We

Table 4 | CoT-decoding is the only decoding strategy that can significantly enhance language models' reasoning.

	GSM8K Acc
Top-k sampling ($k = 10$)	4.9%
Top-p / Nucleus sampling ($p = 0.9$)	6.4%
Beam search ($b = 10$)	6.7%
Temperature sampling ($T = 0.7$)	7.5%
Greedy decoding	9.9%
Self-consistency w/o CoT prompt (10 paths)	12.9%
CoT-decoding ($k = 10$)	**25.1%**

[3]We curate a list of the top 100 celebrity names from (Berglund et al., 2023): https://github.com/lukasberglund/reversal_curse/blob/main/data/celebrity_relations/top_celebrities.txt